Vectors in 3-space

* Representing a 3-space co-ordinate system in 2-space can be a challenge.

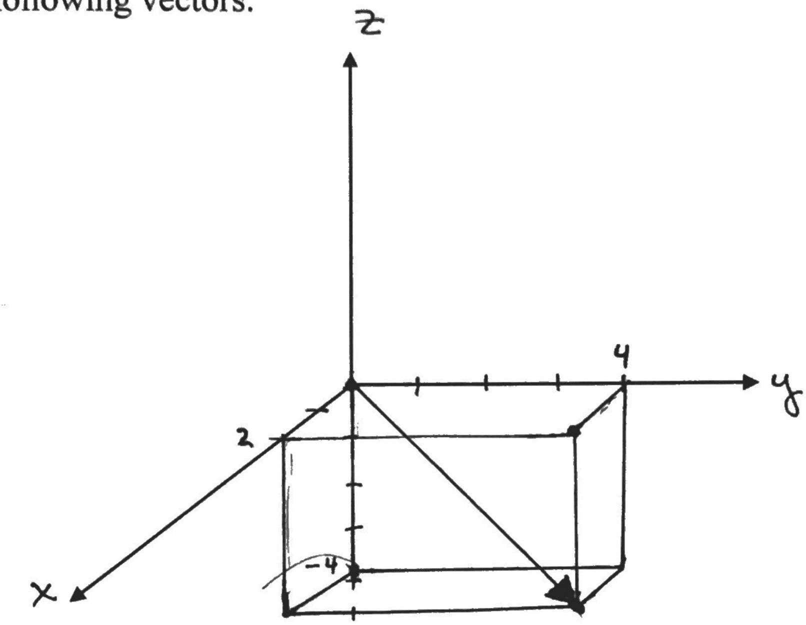
* As a rule, mathematicians use a right-handed co-ordinate system.

* If you grasp the z-axis of a right-handed co-ordinate system with your right hand, pointing your thumb in the direction of the positive z-axis, your fingers should curl from the positive x-axis towards the positive y-axis.

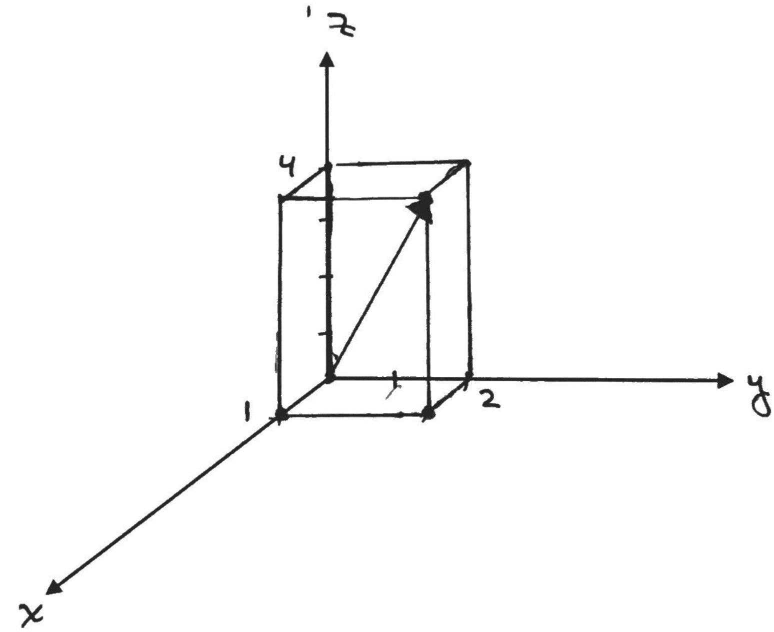
* To plot a point (a, b, c) move "a" units along the x-axis from the origin, then "b" units along the y-axis and then "c" units along the z-axis.

Example: Draw the following vectors.

$[2, 4, -4]$



$[1, 2, 4]$



Notes:

1. Use a rectangular prism to help draw the vector.
2. Plot the *xy* components first and then draw a rectangular prism to determine the location of the *z* component.
3. Draw the vector from the origin to the opposite corner of the prism.
4. 3 space is divided into 8 octants.

Any vector $\vec{u}$ in 3-space can be written as an ordered triple.

* $\vec{u} = \overrightarrow{OP} = [a, b, c]$

* $\vec{u} = a\hat{i} + b\hat{j} + c\hat{k}$

* $\therefore [a, b, c] = a\hat{i} + b\hat{j} + c\hat{k}$

* Magnitude of $\vec{u}$ is $|\vec{u}| = \underbrace{\sqrt{a^2 + b^2 + c^2}}_{\text{follows pythagoras}}$

Vector Addition / Subtraction in 3-space

$\vec{u} = [u_1, u_2, u_3]$ & $\vec{v} = [v_1, v_2, v_3]$, then

$$\vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2, u_3 + v_3]$$

$$\vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2, u_3 - v_3]$$

Scalar Multiplication

$\vec{u} = [u_1, u_2, u_3]$, $k \in \mathbb{R}$, then

$$k\vec{u} = k[u_1, u_2, u_3] = [ku_1, ku_2, ku_3]$$

Vector Between 2 points

vector $\overrightarrow{P_1P_2}$ from $P_1(x_1, y_1, z_1)$ to point $P_2(x_2, y_2, z_2)$ is $\overrightarrow{P_1P_2} = [x_2 - x_1, y_2 - y_1, z_2 - z_1]$

Dot Product for 3-space vectors

$\vec{u} = [u_1, u_2, u_3]$ & $\vec{v} = [v_1, v_2, v_3]$

$$\vec{u} \bullet \vec{v} = u_1v_1 + u_2v_2 + u_3v_3 \quad \Big| \text{ scalar}$$

$$= |\vec{u}| \cdot |\vec{v}| \cos(\theta)$$

Example 1: If $\vec{u} = [2,-3,4]$ & $\vec{v} = [5,-3,-4]$

determine

a) $\vec{u} + \vec{v}$

$= [7,-6,0]$

b) $4\vec{u} - \vec{v}$

$= [3,-9,20]$

Example 2: If $\vec{u} = [2,3,-5]$, $\vec{v} = [8,-4,3]$,

$\vec{w} = [-6,-2,0]$ determine

a) $-3\vec{v}$

$= [-24, 12, -9]$

b) $\vec{u} + \vec{v} + \vec{w}$

$= [4,-3,-2]$

c) $|\vec{u} - \vec{v}|$

$= |[-6, 7, -8]|$

$= \sqrt{(-6)^2 + (7)^2 + (-8)^2}$

$= \sqrt{36 + 49 + 64}$

$= \sqrt{149}$

d) $\vec{u} \bullet \vec{v}$

$= 16 - 12 - 15$

$= -11$

e) $\text{proj}_{\vec{v}}\, \vec{u}$

$= \dfrac{\vec{u} \bullet \vec{v}}{|\vec{v}|^2} \cdot \vec{v}$

$= \dfrac{-11}{89}[8,-4,3]$

$|\vec{v}| = \sqrt{64 + 16 + 9}$

$|\vec{v}| = \sqrt{89}$

Example 3: Determine "b" & "c" such that

$[-2, b, 7]$ and $[c, 6, 21]$ are collinear.

Solution:

Let k represent the scalar.

* Then $[c, 6, 21] = k[-2, b, 7]$

$\therefore$

$c = -2k$

$c = -2(3)$

$c = -6$

$6 = kb$

$b = \frac{6}{k}$

$b = \frac{6}{3}$

$b = 2$

$21 = 7k$

$\therefore k = 3$